

Theorem (Triangle Inequality). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner-product space and $x, y \in V$. Then

$$\|x + y\| \leq \|x\| + \|y\|.$$

Proof. **Step1 (Degenerate case).** If $x = 0$ or $y = 0$ the inequality reduces to $\|y\| \leq \|y\|$ or $\|x\| \leq \|x\|$, which holds with equality. Hence we may assume $x \neq 0$ and $y \neq 0$.

Step2 (Square the inequality). Since norms are non-negative,

$$\|x + y\| \leq \|x\| + \|y\| \iff \|x + y\|^2 \leq (\|x\| + \|y\|)^2.$$

Step3 (Expand both sides).

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2, \end{aligned}$$

and

$$(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\| \|y\| + \|y\|^2.$$

Step4 (Cauchy–Schwarz bound). The Cauchy–Schwarz inequality gives $|\langle x, y \rangle| \leq \|x\| \|y\|$. Since $\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|$,

$$\operatorname{Re} \langle x, y \rangle \leq \|x\| \|y\|.$$

Step5 (Insert the bound). Substituting the estimate from Step4 into the expansion of $\|x + y\|^2$ yields

$$\begin{aligned} \|x + y\|^2 &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2. \end{aligned}$$

Step6 (Square roots). Both sides of the inequality in Step5 are non-negative; taking square roots preserves the order:

$$\|x + y\| \leq \|x\| + \|y\|.$$

Step7 (Equality condition). Equality in the triangle inequality occurs precisely when equality holds in each inequality used above.

- Equality in Cauchy–Schwarz holds iff there exists $\lambda \in \mathbb{C}$ such that $y = \lambda x$.
- Equality $\operatorname{Re} \langle x, y \rangle = |\langle x, y \rangle|$ forces λ to be a non-negative real number; otherwise the real part would be strictly smaller than the modulus.

Thus

$$\|x + y\| = \|x\| + \|y\| \iff (x = 0) \text{ or } (y = 0) \text{ or } (y = \lambda x \text{ with } \lambda \geq 0).$$

In the latter case,

$$\|x + y\| = \|x + \lambda x\| = (1 + \lambda)\|x\| = \|x\| + \|\lambda x\| = \|x\| + \|y\|.$$

Hence the triangle inequality holds for all vectors, and equality holds iff the vectors are positively linearly dependent (including the zero-vector cases). $\square$

$\square$